

18. Ans: A

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

$$V \propto \frac{1}{r} \text{ (since } Q \text{ is constant)}$$

$$\frac{V_P}{V_{surface}} = \frac{r_{surface}}{r_P}$$

$$\begin{aligned} V_P &= \frac{0.10}{0.20} \times (-180) \\ &= -90 \text{ kV} \end{aligned}$$

19. Ans: A